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STAT525 HOMEWORK#6

1. KNNL Problem 9.13 - For part a) generate boxplots or histograms instead of dot plots.
2. KNNL Problem 9.14 - first-order terms mean X_1, X_2 and X_3, second-order terms mean X_i^2, $X_i * X_j$ for all possible i, j.
3. KNNL Problem 9.23.
4. KNNL Problem 9.24.
5. KNNL Problem 10.20. – For part b), please note that $E[Z_{(i)}] \approx \mu + \sigma r_{\alpha_i}$, where $Z_{(i)}$ is the i-th smallest value among $\{Z_1, Z_2, \cdots, Z_n\}$ with each $Z_i \overset{iid}{\sim} N(\mu, \sigma^2)$, $\alpha_i = \frac{i-3/8}{n+1/4}$, and r_{α_i} is the $\alpha_i \times 100\%$ percentile of $N(0,1)$. You can also use `PROC RANK` to output the expected values.
6. KNNL Problem 10.24.
7. KNNL Problem 10.26.

1. KNNL Problem 9.13 - For part a) generate boxplots or histograms instead of dot plots.

9.13. Lung pressure. Increased arterial blood pressure in the lungs frequently leads to the development of heart failure in patients with chronic obstructive pulmonary disease (COPD). The standard method for determining arterial lung pressure is invasive, technically difficult, and involves some risk to the patient. Radionuclide imaging is a noninvasive, less risky method for estimating arterial pressure in the lungs. To investigate the predictive ability of this method, a cardiologist collected data on 19 mild-to-moderate COPD patients. The data that follow on the next page include the invasive measure of systolic pulmonary arterial pressure (Y) and three potential noninvasive predictor variables. Two were obtained by using radionuclide imaging—emptying rate of blood into the pumping chamber of the heart (X_1) and ejection rate of blood pumped out of the heart into the lungs (X_2)—and the third predictor variable measures a blood gas (X_3).

a. Prepare separate dot plots for each of the three predictor variables. Are there any noteworthy features in these plots? Comment.

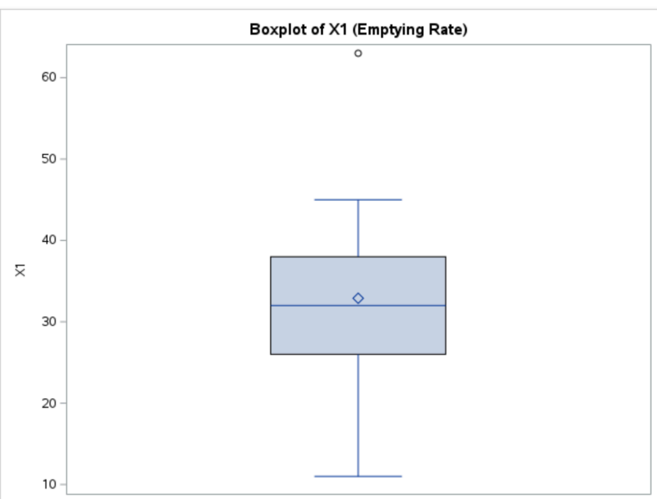
Subject i	X_{i1}	X_{i2}	X_{i3}	Y_i
1	45	36	45	49
2	30	28	40	55
3	11	16	42	85
...	...	...	...	...
17	27	51	44	29
18	37	32	54	40
19	34	40	36	31

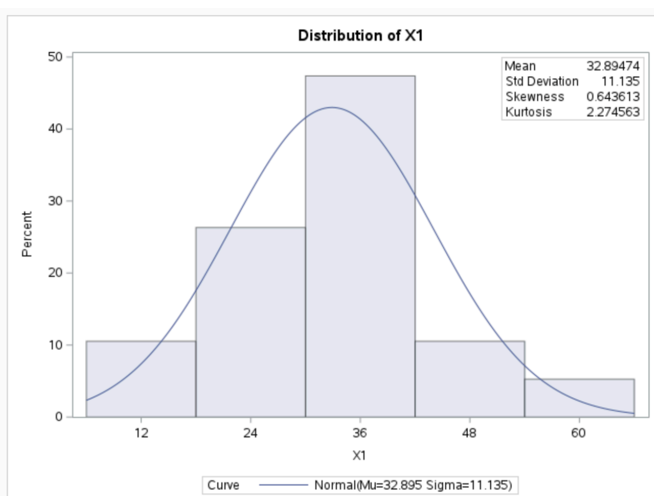
Adapted from A. T. Marmor et al., "Improved Radionuclide Method for Assessment of Pulmonary Artery Pressure in COPD," *Chest* 89 (1986), pp. 64–69.

```
data lungpressure;
    input Y X1 X2 X3;
    datalines;
  49.0    45.0    36.0    45.0
  55.0    30.0    28.0    40.0
  85.0    11.0    16.0    42.0
  32.0    30.0    46.0    40.0
  26.0    39.0    76.0    43.0
  28.0    42.0    78.0    27.0
  95.0    17.0    24.0    36.0
  26.0    63.0    80.0    42.0
  74.0    25.0    12.0    52.0
  37.0    32.0    27.0    35.0
  31.0    37.0    37.0    55.0
  49.0    29.0    34.0    47.0
  38.0    26.0    32.0    28.0
  41.0    38.0    45.0    30.0
  12.0    38.0    99.0    26.0
  44.0    25.0    38.0    47.0
  29.0    27.0    51.0    44.0
  40.0    37.0    32.0    54.0
  31.0    34.0    40.0    36.0
;
run;
```

Y   X1   X2   X3

```
proc sgplot data=lungpressure;
    vbox X1;
    title "Boxplot of X1 (Emptying Rate)";
run;

proc sgplot data=lungpressure;
    vbox X2;
    title "Boxplot of X2 (Ejection Rate)";
run;

proc sgplot data=lungpressure;
    vbox X3;
    title "Boxplot of X3 (Blood Gas Variable)";
run;

proc univariate data=lungpressure normal;
    var X1 X2 X3;
    histogram X1 X2 X3 / normal;
    inset mean std skewness kurtosis / position=ne;
    title "Histograms of X1, X2, X3";
run;
quit;
```

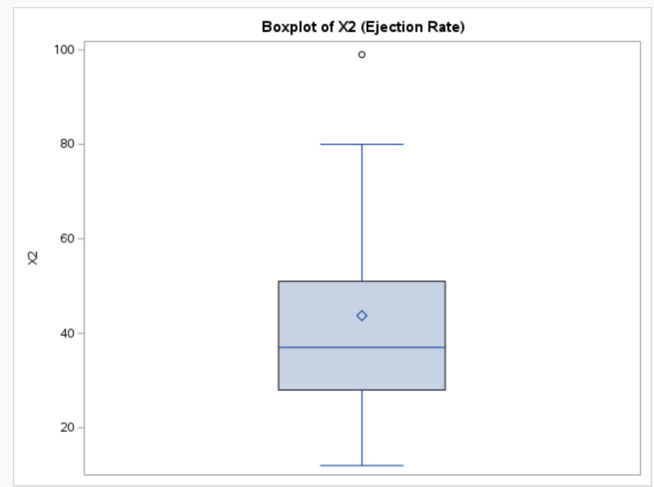


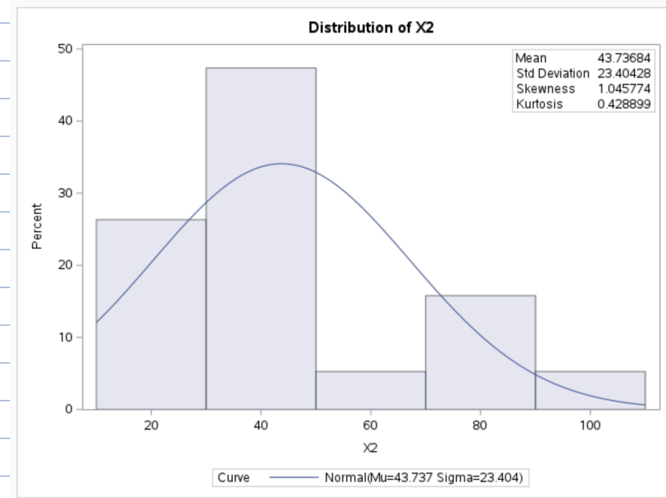


X_1 (Emptying Rate)

- Distribution is roughly symmetric with mean ≈ 33.
- One potential outlier near 60.
- Skewness ≈ 0.64, slightly right-skewed.

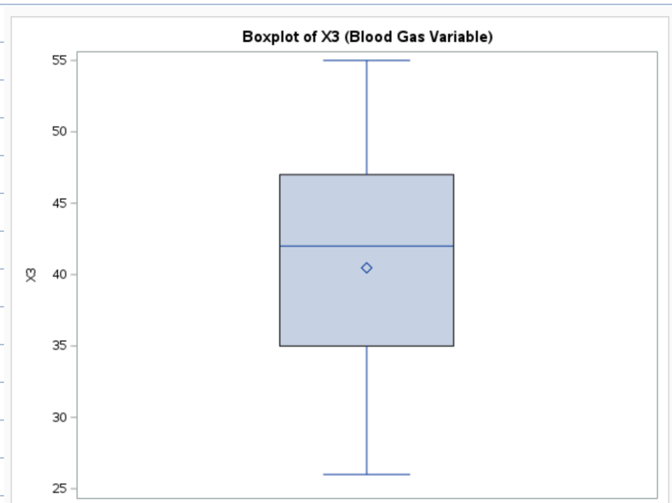
→ ◦ X_1 has one moderate high value but otherwise well-behaved.

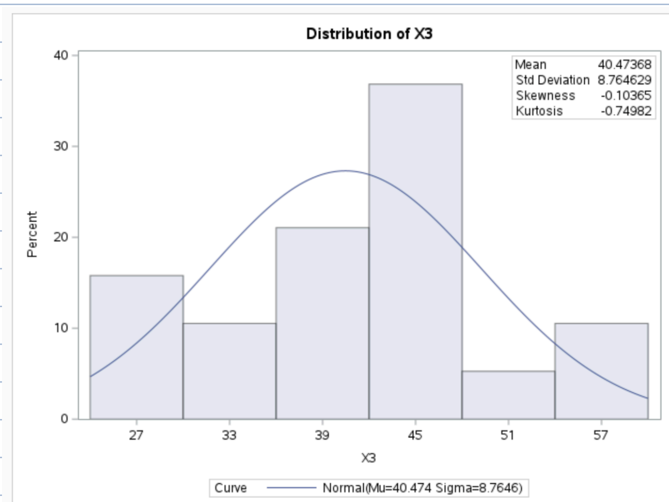




X_1 (Ejection Rate)

- Distribution showing unsymmetry with mean ≈ 44, SD ≈ 23.
- One potential outlier near 100.
- Skewness ≈ 1.05, clearly right-skwed.
- Values cluster at around 30-50, with a few that stretched upper tail as well.
- → X_2 shows the strongest skewness and outlier.





X_3 (Blood Gas)

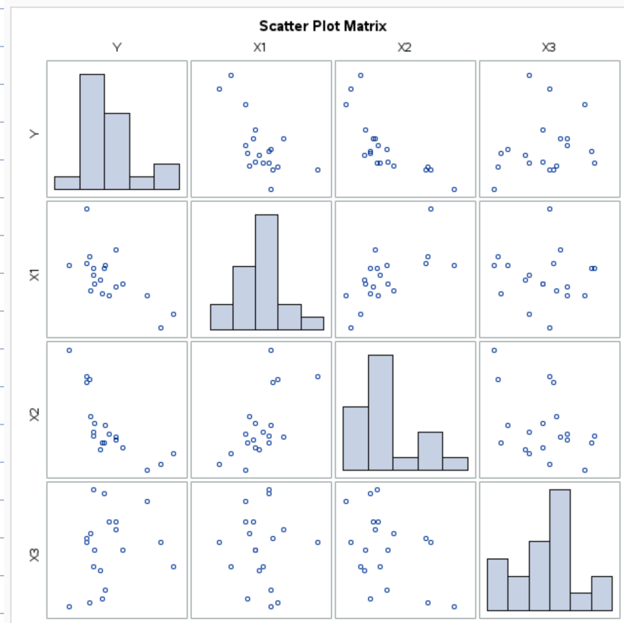
- Distribution is approx. symmetric with mean ≈ 40, skewness ≈ -0.10.
- No potential outlier observed, range 26-55.
- Variation smaller than X_1 and X_2.
- → X_3 appears the most normal + balanced.

b. Obtain the scatter plot matrix. Also obtain the correlation matrix of the X variables. What do the scatter plots suggest about the nature of the functional relationship between Y and each of the predictor variables? Are any serious multicollinearity problems evident? Explain.

```
proc sgscatter data=lungpressure;
    matrix Y X1 X2 X3 / diagonal=(histogram);
    title "Scatterplot Matrix for Y, X1, X2, and X3";
run;

proc corr data=lungpressure plots=matrix(histogram);
    var Y X1 X2 X3;
    title "Correlation Matrix of Y and X Variables";
run;
quit;
```

Pearson Correlation Coefficients, N = 19
Prob > |r| under H0: Rho=0

	Y	X1	X2	X3
Y	1.00000	-0.66505 0.0019	-0.74757 0.0002	0.22387 0.3569
X1	-0.66505 0.0019	1.00000	0.65285 0.0024	-0.04614 0.8512
X2	-0.74757 0.0002	0.65285 0.0024	1.00000	-0.42348 0.0708
X3	0.22387 0.3569	-0.04614 0.8512	-0.42348 0.0708	1.00000



- Y v.s. X_1 } • Both negatively associated with Y.
  Y v.s. X_2 } (p-values both significant)
- Y v.s. X_3 } Positively associated with Y.
  (However, p-value not significant)
- Multicollinearity of X_1 v.s. X_2 ($r \approx 0.652$, $p < 0.05$), but not too severe.
- All relationships appear roughly linear, meaning multiple linear regression is appropriate to use.

c. Fit the multiple regression function containing the three predictor variables as first-order terms. Does it appear that all predictor variables should be retained?

```
proc reg data=lungpressure;
    model Y = X1 X2 X3 / vif tol;
    title "Multiple Regression Model for Lung Pressure
run;
quit;
```

Multiple Regression Model for Lung Pressure (Y = X1 + X2 + X3)

The REG Procedure
Model: MODEL1
Dependent Variable: Y

Number of Observations Read	19
Number of Observations Used	19

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	4966.67801	1655.55934	7.96	0.0021
Error	15	3121.00620	208.06708		
Corrected Total	18	8087.68421			

Root MSE	14.42453	R-Square	0.6141
Dependent Mean	43.26316	Adj R-Sq	0.5369
Coeff Var	33.34137		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	87.18750	21.55246	4.05	0.0011	.	0
X1	1	-0.56448	0.42791	-1.32	0.2069	0.50914	1.96410
X2	1	-0.51315	0.22449	-2.29	0.0372	0.41872	2.38821
X3	1	-0.07196	0.45457	-0.16	0.8763	0.72820	1.37324

$$\hat{Y} = 87.19 - 0.56X_1 - 0.51X_2 - 0.07X_3$$

- VIF < 10, Tol > 0.1 → no serious multicollinearity. (Keep both X_1+X_2)
- X_1 Border line significant, X_2 significant, X_3 not significant.
- Therefore, it is reasonable to drop X_3 for the next step.
- Keep X_1 for now as X_1 is still significantly correlated with Y, and VIF and Tol doesn't show X_1 is harming the model.

2. KNNL Problem 9.14 - first-order terms mean X_1, X_2 and X_3, second-order terms mean X_i^2, $X_i * X_j$ for all possible i, j.

9.14. Refer to **Lung pressure** Problem 9.13.

a. Using first-order and second-order terms for each of the three predictor variables (centered around the mean) in the pool of potential X variables (including cross products of the first-order terms), find the three best hierarchical subset regression models according to the $R^2_{a,p}$ criterion.

First Order: Original Predictor, linear terms

X_1, X_2, X_3

Second Order: ① Squared Terms, X_i^2

② Cross Product (Interaction) Terms, X_iX_j

$X_1^2, X_2^2, X_3^2, X_1X_2, X_1X_3, X_2X_3$

```
/* Step 1: compute means and store as macro variables */
proc means data=lungpressure noprint;
    var X1 X2 X3;
    output out=means mean=meanX1 meanX2 meanX3;
run;

/* Step 2: center variables using those means */
data lungpressure_centered;
    if _n_ = 1 then set means;
    set lungpressure;
    x1c = X1 - meanX1;
    x2c = X2 - meanX2;
    x3c = X3 - meanX3;
run;

/* Step 3: generate second-order terms */
data lungpressure2;
    set lungpressure_centered;
    x1sq = x1c*x1c;
    x2sq = x2c*x2c;
    x3sq = x3c*x3c;
    x1x2 = x1c*x2c;
    x1x3 = x1c*x3c;
    x2x3 = x2c*x3c;
run;

/* Step 4: run all-subset selection on first- and second-order terms */
proc reg data=lungpressure2;
    model Y = x1c x2c x3c x1sq x2sq x3sq x1x2 x1x3 x2x3 /
              selection=adjrsq best=3;
    title "Best Three Hierarchical Subset Models by Adjusted R-Square";
run;
```

Best Three Hierarchical Subset Models by Adjusted R-Square

The REG Procedure
Model: MODEL1
Dependent Variable: Y

Adjusted R-Square Selection Method

Number of Observations Read	19
Number of Observations Used	19

Number in Model	Adjusted R-Square	R-Square	Variables in Model
4	0.7507	0.8061	x1c x2c x1sq x2sq
3	0.7507	0.7922	x1c x2c x1x2
4	0.7458	0.8023	x1c x2c x1sq x2x3

- All 3 models include X_1 centered and X_2 centered, and at least one Second Order term.

b. Is there much difference in $R^2_{a,p}$ for the three best subset models?

- No much difference in the R^2_{adj} for all 3 models, so all 3 models fit equally well.
- Meaning that adding a Second Order term does not improve fit that much.

3. KNNL Problem 9.23.

9.23. Refer to **Lung pressure** Problems 9.13 and 9.14. The validity of the regression model identified as best in Problem 9.14a is to be assessed internally.

Best Model:
X_{1C} X_{2C} X_1X_2

a. Calculate the *PRESS* statistic and compare it to *SSE*. What does this comparison suggest about the validity of *MSE* as an indicator of the predictive ability of the fitted model?

```
proc reg data=lungpressure2 outest=stats press;
    model Y = x1c x2c x1x2;
run;

proc print data=stats; run;
```

Obs	_MODEL_	_TYPE_	_DEPVAR_	_RMSE_	_PRESS_	Intercept	x1c	x2c	x1x2	Y
1	MODEL1	PARMS	Y	10.5845	5102.49	37.8882	-0.67453	-0.60239	0.033347	-1

The REG Procedure
Model: MODEL1
Dependent Variable: Y

Number of Observations Read	19
Number of Observations Used	19

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	6407.21941	2135.73980	19.06	<.0001
Error	15	1680.46480	112.03099		
Corrected Total	18	8087.68421			

Root MSE	10.58447	R-Square	0.7922
Dependent Mean	43.26316	Adj R-Sq	0.7507
Coeff Var	24.46532		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	37.88818	2.85220	13.28	<.0001
x1c	1	-0.67453	0.29678	-2.27	0.0382
x2c	1	-0.60239	0.14388	-4.19	0.0008
x1x2	1	0.03335	0.00928	3.59	0.0027

PRESS: 5102.49
SSE: 1680.46
MSE: 112.03

PRESS >>> SSE : Bad

- Indicating fitted model predicts new observation less accurately than it fits the existing data.
- MSE based on SSE is not a reliable indicator of predictive ability in this model, possibly by overfitting or limited generalizability.

Ideally:
- SSE: ↓↓↓
- PRESS: close to SSE, ↓↓↓
- PRESS/SSE ≤ 1.5, Acceptable Ratio

b. Case 8 alone accounts for approximately one-half of the entire *PRESS* statistic. Would you recommend modification of the model because of the strong impact of this case? What are some corrective action options that would lessen the effect of case 8? Discuss.

- Case 8 contributes to half of PRESS Statistic, meaning it is a influential observation.
- Case 8 strongly affects the model's predictive power.
- Instead of immediately modifying the model based solely in this information, we should first verify data accuracy and examine its leverage and the studentized residual.
- This is because if Case 8 is extreme but still valid, we should opt for ↑ robust regression, or transforming the variable to reduce the observations influence.
- Either way, Case 8's affect shows the fitted model has limited stability for atypical observations.

Mallow's C_p Criterion

4. KNNL Problem 9.24.

9.24 The true quadratic regression function is $E\{Y\} = 15 + 20X + 3X^2$. The fitted linear regression function is $\hat{Y} = 13 + 40X$, for which $E\{b_0\} = 10$ and $E\{b_1\} = 45$. What are the bias and sampling error components of the mean squared error for $X_i = 10$ and for $X_i = 20$?

- $\hat{Y} = 13 + 40X$ is just one sample line from a dataset.
- $E(b_0) = 10$ ✓ } Expected values of the coefficient over repeated samples. $\xrightarrow{\text{use } E(\hat{Y})}$ $E(\hat{Y}) = 10 + 45X$ ✓
- $E(b_1) = 45$ ✓

For $X_i = 10$:

$E(\hat{Y}) = 10 + 45(10) = 460$

$E(Y) = 15 + 20(10) + 3(10)^2 = 515$

For $X_i = 20$:

$E(\hat{Y}) = 10 + 45(20) = 910$

$E(Y) = 15 + 20(20) + 3(20)^2 = 1615$

$\boxed{\text{Bias} = E(\hat{Y}) - E(Y)}$

- Bias $= 460 - 515 = -55$
- $\text{Bias}^2 = 3025$

- Bias $= 910 - 1615 = -705$
- $\text{Bias}^2 = 497025$

$$\boxed{MSE(\hat{Y}_i) = \underbrace{[E(\hat{Y}_i) - E(Y_i)]^2}_{\text{Bias}^2} + \underbrace{Var(\hat{Y}_i)}_{\text{Sampling Error}}}$$

5. KNNL Problem 10.20. – For part b), please note that $E[Z_{(i)}] \approx \mu + \sigma r_{\alpha_i}$, where $Z_{(i)}$ is the i-th smallest value among $\{Z_1, Z_2, \cdots, Z_n\}$ with each $Z_i \overset{iid}{\sim} N(\mu, \sigma^2)$, $\alpha_i = \frac{i-3/8}{n+1/4}$, and r_{α_i} is the $\alpha_i \times 100\%$ percentile of $N(0,1)$. You can also use PROC RANK to output the expected values.

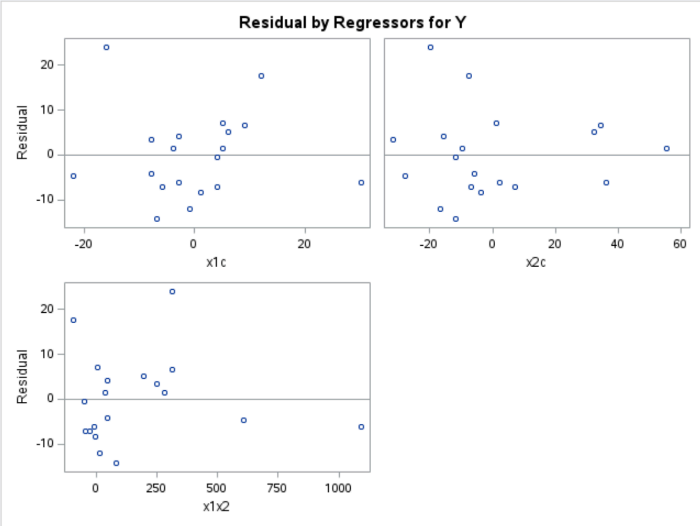
10.20. Refer to **Lung pressure** Problems 9.13 and 9.14. The subset regression model containing first-order terms for X_1 and X_2 and the cross-product term X_1X_2 is to be evaluated in detail.

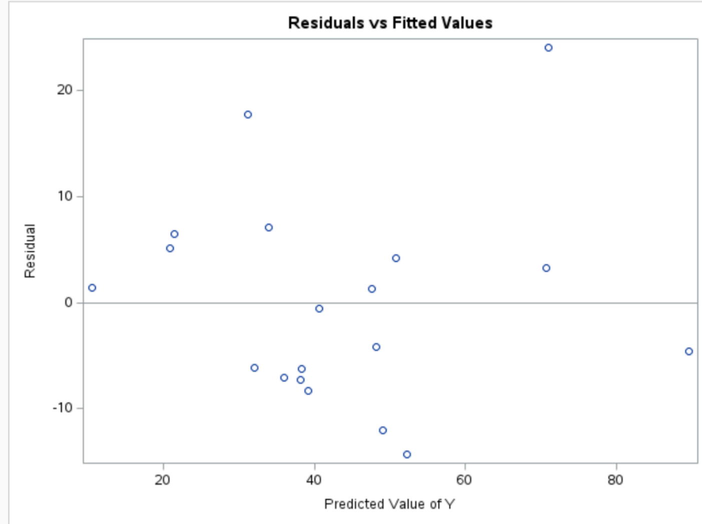
$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \cdot X_2)$

a. Obtain the residuals and plot them separately against $\hat{Y}$ and each of the three predictor variables. On the basis of these plots, should any further modifications of the regression model be attempted?

```sas
/* Fit the chosen subset model and output residuals */
proc reg data=lungpressure2;
    model Y = x1c x2c x1x2;
    output out=residuals r=resid p=fitted;
run;
quit;

/* Plot residuals vs fitted values */
proc sgplot data=residuals;
    scatter x=fitted y=resid;
    refline 0 / axis=y;
    title "Residuals vs Fitted Values";
run;
```

```sas
/* Residuals vs x1c */
proc sgplot data=residuals;
    scatter x=x1c y=resid;
    refline 0 / axis=y;
    title "Residuals vs x1c";
run;

/* Residuals vs x2c */
proc sgplot data=residuals;
    scatter x=x2c y=resid;
    refline 0 / axis=y;
    title "Residuals vs x2c";
run;

/* Residuals vs interaction term */
proc sgplot data=residuals;
    scatter x=x1x2 y=resid;
    refline 0 / axis=y;
    title "Residuals vs x1x2";
run;
```





- Residual Plots show no major violation of linearity, but there's evidence of unequal variance (↑ spread of e_i as ↑ predictor value).

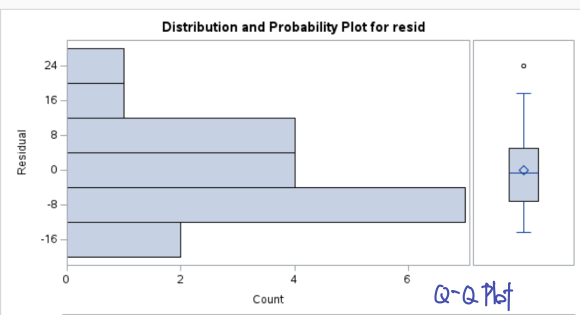
- No additional polynomial terms should be added at the moment, but should check for potential variance stabilization method (WLS).

- Current model is well, with some heteroscedasticity.

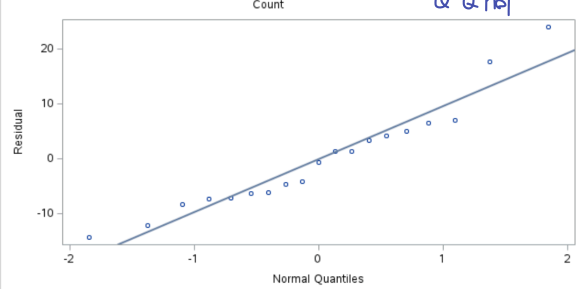
Q-Q

b. Prepare a normal probability plot of the residuals. Also obtain the coefficient of correlation between the ordered residuals and their expected values under normality. Does the normality assumption appear to be reasonable here?

```
proc univariate data=residuals normal plot;
    var resid;
    qqplot resid / normal(mu=est sigma=est);
run;
```



Q-Q Plot



The UNIVARIATE Procedure
Variable: resid (Residual)

Moments			
N	19	Sum Weights	19
Mean	0	Sum Observations	0
Std Deviation	9.66225417	Variance	93.3591557
Skewness	0.94434884	Kurtosis	1.0143378
Uncorrected SS	1680.4648	Corrected SS	1680.4648
Coeff Variation	.	Std Error Mean	2.21667313

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	9.66225
Median	-0.58243	Variance	93.35916
Mode	.	Range	38.34732
		Interquartile Range	12.15770

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	-0.5	Pr >= |M|	1.0000
Signed Rank	S	-12	Pr >= |S|	0.6507

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.93228	Pr < W	0.1908
Kolmogorov-Smirnov	D	0.141017	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.065467	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.463626	Pr > A-Sq	0.2337

- Q-Q Plot is approx. linear with only minor right-tail deviation. ✓
- Normality assumption is reasonable for this model.

- Fail to Reject H_0
- All normality tests fail to Reject H_0, confirming normality assumption is reasonable for this model.

c. Obtain the variance inflation factors. Are there any indications that serious multicollinearity problems are present? Explain.

```
proc reg data=lungpressure2;
    model Y = x1c x2c x1x2 / vif tol;
    title "Variance Inflation Factors for X1c, X2c, and X1x2";
run;
```

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Tolerance	Variance Inflation
Intercept	1	37.88818	2.85220	13.28	<.0001	.	0
x1c	1	-0.67453	0.29678	-2.27	0.0382	0.56994	1.75458
x2c	1	-0.60239	0.14388	-4.19	0.0008	0.54887	1.82193
x1x2	1	0.03335	0.00928	3.59	0.0027	0.88439	1.13073

- VIF < 10, Tol > 0.1 → no serious multicollinearity.
- Predictors are sufficiently independent and stable enough for prediction.

d. Obtain the studentized deleted residuals and identify any outlying Y observations. Use the Bonferroni outlier test procedure with $\alpha = .05$. State the decision rule and conclusion.

```
proc reg data=lungpressure2;
    model Y = x1c x2c x1x2 / influence;
    output out=outres rstudent=rstud;
    title "Studentized Deleted Residuals and Bonferroni Outlier Test";
run;

proc print data=outres;
    var Y rstud;
run;
quit;
```

Studentized Deleted Residuals and Bonferroni Outlier Test

The REG Procedure
Model: MODEL1
Dependent Variable: Y

Output Statistics									
						DFBETAS			
Obs	Residual	RStudent	Hat Diag H	Cov Ratio	DFFITS	Intercept	x1c	x2c	x1x2
1	17.7397	2.2095	0.2757	0.5499	1.3632	0.8195	1.0978	-0.6979	-0.5955
2	4.1605	0.3989	0.0834	1.3741	0.1203	0.0923	0.0248	-0.0594	-0.0209
3	-4.6164	-0.6292	0.5389	2.5561	-0.6802	0.0718	0.3822	0.0726	-0.4819
4	-6.2590	-0.6049	0.0848	1.2988	-0.1842	-0.1722	0.0558	-0.0706	0.0929
5	5.0963	0.5172	0.1757	1.4821	0.2387	0.1334	-0.0587	0.1857	-0.0422
6	6.4898	0.6618	0.1737	1.4100	0.3035	0.1367	-0.0320	0.2039	0.0104
7	24.0398	3.3141	0.2178	0.1661	1.7486	0.2873	-1.0884	-0.0756	0.8475
8	-6.1424	-1.7794	0.8783	4.7895	-4.7798	0.7275	-2.4073	1.0195	-3.2858
9	3.3135	0.3379	0.1925	1.5799	0.1650	0.0367	0.0148	-0.1168	0.0701
10	-12.0731	-1.2232	0.1017	0.9773	-0.4116	-0.3054	-0.1630	0.2391	0.1017
11	-7.2550	-0.7153	0.1116	1.2849	-0.2534	-0.2103	-0.1407	0.0974	0.1183
12	1.3548	0.1282	0.0680	1.4073	0.0346	0.0309	-0.0024	-0.0056	-0.0095
13	-14.3075	-1.4574	0.0753	0.8100	-0.4159	-0.3130	0.1345	0.0381	0.0323
14	7.1013	0.6921	0.0929	1.2699	0.2215	0.1995	0.1044	-0.0314	-0.1097
15	1.4369	0.1821	0.4798	2.5095	0.1749	0.0575	-0.0835	0.1617	-0.0157
16	-4.1795	-0.4021	0.0897	1.3826	-0.1262	-0.0972	0.0694	-0.0338	0.0283
17	-7.0614	-0.7092	0.1444	1.3375	-0.2914	-0.2245	0.1624	-0.1828	0.1426
18	-0.5824	-0.0573	0.1391	1.5292	-0.0230	-0.0172	-0.0141	0.0119	0.0099
19	-8.2559	-0.8021	0.0768	1.1926	-0.2314	-0.2230	-0.0620	0.0332	0.1143

Sum of Residuals	0
Sum of Squared Residuals	1680.46480
Predicted Residual SS (PRESS)	5102.49435

- Maximum |RStud|: obs #7, 3.3141
- Second ↑ |RStud|: obs #1, 2.2095

$\boxed{\text{Bonferroni-adjusted } \alpha\text{-level}: \alpha/2n}$

- Bonferroni-adjusted α-level: $0.05/2 \cdot 19 = 0.0013$ (two-sided)
- Critical t ($\boxed{df = (n-1)-p-1} = (19-1)-3-1 = 14$)

⇓

Decision Rule: Flag as outlier if $p < 0.0013 = \alpha^*$

- Maximum |RStud|: obs #7, 3.3141

  $p: 2(1 - F_t(3.3141; 14)) \approx 0.005 > 0.0013 = \alpha^*$

Conclusion: No outlying Y observations detected at $\alpha = 0.05$ using the Bonferroni outlier test.

e. Obtain the diagonal elements of the hat matrix. Using the rule of thumb in the text, identify any outlying X observations. Are your findings consistent with those in Problem 9.13a? Should they be? Discuss.

Result in 5(d) for H Diag.

$\boxed{\text{Rule of } h_{ii} > 2p/n}$ $\quad h_{ii} = 2(3)/19 \approx 0.316$

4 obs w/ ↑ leverage:
- obs # 3
- obs # 6
- obs # 7
- obs # 12

- Results are consistent with 9.13(a), where obs. with extreme X values were identified earlier from scatterplots.

f. Cases 3, 8, and 15 are moderately far outlying with respect to their X values, and case 7 is relatively far outlying with respect to its Y value. Obtain *DFFITS*, *DFBETAS*, and Cook's distance values for these cases to assess their influence. What do you conclude?

Result in 5(d) for DFFITS and DFBETAS.

```
proc reg data=lungpressure2;
    model Y = x1c x2c x1x2 / influence r;
    output out=influence_out cookd=D dffits=dffits
           dfbetas_intercept=dfb_int dfbetas_x1c=dfb_x1c
           dfbetas_x2c=dfb_x2c dfbetas_x1x2=dfb_x1x2;
    title "Influence Diagnostics: DFFITS, DFBETAS, and Cook's D";
run;

proc print data=influence_out;
    var Y dffits D dfb_int dfb_x1c dfb_x2c dfb_x1x2;
    title "Influence Output for All Observations";
run;
```

Influence Diagnostics: DFFITS, DFBETAS, and Cook's D

The REG Procedure
Model: MODEL1
Dependent Variable: Y

Output Statistics

Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D	RStudent	Hat Diag H	Cov Ratio	DFFITS	DFBETAS Intercept	DFBETAS x1c	DFBETAS x2c	DFBETAS x1x2
1	49	39.3334	2.3022	9.6666	4.083	2.368	0.446	2.8902	0.2412	0.2672	1.6296	1.4813	1.3645	-1.2004	-1.2855
2	55	54.7015	2.0074	0.2985	4.236	0.070	0.000	0.0681	0.1834	1.6117	0.0323	0.0213	-0.0029	0.0132	-0.0136
3	85	86.3668	2.8845	-1.3668	3.695	-0.370	0.021	-0.3590	0.3787	2.0447	-0.2803	0.0406	0.0909	-0.1188	-0.1335
4	32	24.7426	2.0289	7.2574	4.225	1.718	0.170	1.8513	0.1874	0.6754	0.8889	-0.1928	-0.4785	0.1998	0.6144
5	26	24.1571	1.8983	1.8429	4.286	0.430	0.009	0.4180	0.1640	1.5001	0.1852	0.0129	0.0026	-0.0645	0.0658
6	28	28.6547	1.4184	-0.6547	4.467	-0.147	0.001	-0.1417	0.0916	1.4424	-0.0450	-0.0134	0.0017	0.0097	-0.0097
7	95	92.7625	3.5440	2.2375	3.068	0.729	0.178	0.7175	0.5717	2.6629	0.8289	-0.2016	-0.1437	0.1462	0.4534
8	26	35.9791	1.2132	-9.9791	4.528	-2.204	0.087	-2.5896	0.0670	0.2952	-0.6939	-0.5118	0.0025	-0.0207	0.1681
9	74	76.4629	3.4810	-2.4629	3.139	-0.785	0.189	-0.7741	0.5515	2.4850	-0.8584	-0.5734	-0.7303	0.5740	0.5207
10	37	41.0824	2.0736	-4.0824	4.204	-0.971	0.057	-0.9692	0.1957	1.2636	-0.4781	-0.1015	0.3150	-0.3724	-0.0654
11	31	29.2377	1.7242	1.7623	4.359	0.404	0.006	0.3928	0.1353	1.4587	0.1554	-0.0116	-0.1037	0.0718	0.0907
12	49	47.2963	1.6992	1.7037	4.368	0.390	0.006	0.3787	0.1314	1.4566	0.1473	0.1337	0.0371	0.0041	-0.0968
13	38	36.5596	1.6685	1.4404	4.380	0.329	0.004	0.3188	0.1267	1.4660	0.1215	0.0266	-0.0787	0.0820	0.0276
14	41	41.6276	1.4314	-0.6276	4.463	-0.141	0.001	-0.1359	0.0933	1.4457	-0.0436	-0.0397	-0.0090	0.0004	0.0248
15	12	16.2875	3.2584	-4.2875	3.369	-1.272	0.379	-1.3016	0.4833	1.6149	-1.2587	0.0434	-0.2968	0.7362	-0.3808
16	44	41.6276	1.4314	2.3724	4.463	0.532	0.007	0.5184	0.0933	1.3469	0.1663	0.1515	0.0341	-0.0014	-0.0946
17	29	32.5769	1.5545	-3.5769	4.422	-0.809	0.020	-0.7991	0.1100	1.2387	-0.2809	-0.2164	-0.1633	0.1837	0.1227
18	40	39.3813	1.6707	0.6187	4.379	0.141	0.001	0.1366	0.1270	1.5016	0.0521	0.0185	-0.0284	0.0340	0.0028
19	31	33.1624	1.2180	-2.1624	4.526	-0.478	0.004	-0.4651	0.0675	1.3292	-0.1252	-0.0643	0.0221	-0.0104	-0.0050

Based on the DFFITS, DFBETAS, and Cook's distance values from the model:

- Observation 8 shows strong influence on the regression model. It has a large DFFITS value (greater than the cutoff) and moderate Cook's D, indicating that removing this observation would noticeably change the fitted values.

- Observation 15 shows moderate influence, particularly on individual regression coefficients (its DFBETAS values exceed ±1 for some predictors). It should be examined for data quality or model fit problems.

- Observations 3 and 7, although somewhat outlying in X or Y respectively, do not show strong influence based on standard diagnostics and are not considered influential.

Conclusion:
Observation 8 is clearly influential, 15 is moderately influential, and 3 and 7 are not influential but may be worth keeping an eye on.

6. KNNL Problem 10.24.

10.24. If $n = p$ and the $\mathbf{X}$ matrix is invertible, use (5.34) and (5.37) to show that the hat matrix $\mathbf{H}$ is given by the $p \times p$ identity matrix. In this case, what are h_{ii} and $\hat{Y}_i$?

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1} \qquad (5.34)$$

$$(\mathbf{A}')^{-1} = (\mathbf{A}^{-1})' \qquad (5.37)$$

Step 1: Start with Hat Matrix Formula

The hat matrix is:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

Since X is a **square invertible matrix** ($n = p$), we can use the matrix inverse identity:

$$(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{X}^{-1}(\mathbf{X}')^{-1}$$

(from equation 5.34)

Step 2: Substitute Into H

$$\mathbf{H} = \mathbf{X}\left(\mathbf{X}^{-1}(\mathbf{X}')^{-1}\right)\mathbf{X}'$$

Rearrange:

$$\mathbf{H} = \left(\mathbf{X}\mathbf{X}^{-1}\right)\left((\mathbf{X}')^{-1}\mathbf{X}'\right)$$

$$\mathbf{H} = \mathbf{I}\mathbf{I} = \mathbf{I}$$

So:

$$\boxed{\mathbf{H} = \mathbf{I}}$$

Step 3: Answer the Question

Since $\mathbf{H} = \mathbf{I}$:

- The diagonal elements of H, h_{ii}, are:

$$\boxed{h_{ii} = 1 \text{ for all } i}$$

- The fitted values:

$$\hat{\mathbf{Y}} = \mathbf{HY} = \mathbf{IY} = \mathbf{Y}$$

So:

$$\boxed{\hat{Y}_i = Y_i \text{ for all } i}$$

When n=p, the regression model perfectly fits the data—there is no residual error. The hat matrix becomes the identity matrix, each observation has maximum leverage (1), and the predicted values equal the observed values.

7. KNNL Problem 10.26.

10.26. Prove (9.11), using (10.27) and Exercise 5.31.

To see why C_p as defined in (9.9) is an estimator of Γ_p, we need to utilize two results that we shall simply state. First, it can be shown that:

$$\sum_{i=1}^{n} \sigma^2\{\hat{Y}_i\} = p\sigma^2 \tag{9.11}$$

The diagonal elements h_{ii} of the hat matrix have some useful properties. In particular, their values are always between 0 and 1 and their sum is p:

$$0 \leq h_{ii} \leq 1 \qquad \sum_{i=1}^{n} h_{ii} = p \tag{10.27}$$

5.31. Obtain an expression for the variance-covariance matrix of the fitted values $\hat{Y}_i$, $i = 1, \ldots, n$, in terms of the hat matrix.

Step 1: Use Result from Exercise 5.31

Exercise **5.31** tells us that the **variance–covariance matrix of the fitted values** is:

$$\operatorname{Var}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}$$

where $\mathbf{H}$ is the hat matrix.

Step 2: Diagonal Elements of $\operatorname{Var}(\hat{Y}_i)$

From this result, the variance of each fitted value is:

$$\operatorname{Var}(\hat{Y}_i) = \sigma^2 h_{ii}$$

because the diagonal elements of $\sigma^2 H$ are $\sigma^2 h_{ii}$.

Step 3: Use Result (10.27)

Result **(10.27)** tells us:

$$\sum_{i=1}^{n} h_{ii} = p$$

where p is the number of parameters (including the intercept) in the regression model.

Step 4: Combine Results

Now sum the variances of the fitted values:

$$\sum_{i=1}^{n} \operatorname{Var}(\hat{Y}_i) = \sum_{i=1}^{n} \sigma^2 h_{ii} = \sigma^2 \sum_{i=1}^{n} h_{ii} = \sigma^2 p$$

$$\boxed{\sum_{i=1}^{n} \operatorname{Var}(\hat{Y}_i) = p\sigma^2}$$

STAT525 HOMEWORK#6 Solution

1. KNNL Problem 9.13 - For part a) generate boxplots or histograms instead of dot plots.

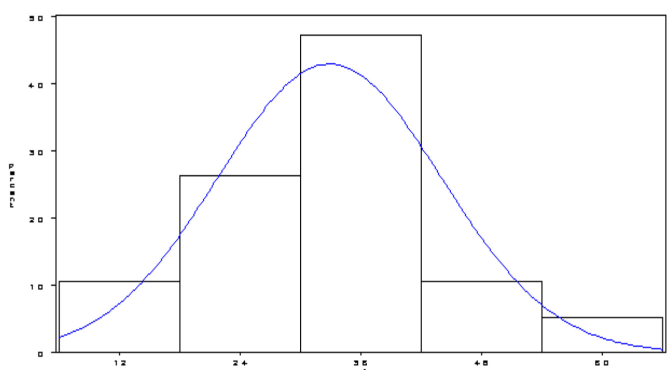
   The histograms reveal that variables x2 and x3 are a bit skewed but there is nothing too alarming in the plots. As far as the scatterplot matrix, the strongest linear relationship is between x1 and x2 followed by x2 and y. There does not appear to be any critical pairwise correlation. Perhaps a few influential points. The model output is shown below. It does not appear that all variables need to be included.

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	4966.67801	1655.55934	7.96	0.0021
Error	15	3121.00620	208.06708		
Corrected Total	18	8087.68421			

Root MSE	14.42453	R-Square	0.6141
Dependent Mean	43.26316	Adj R-Sq	0.5369
Coeff Var	33.34137		

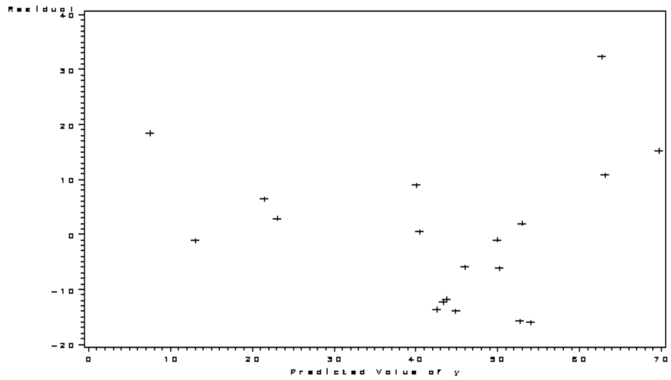
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	87.18750	21.55246	4.05	0.0011	0
x1	1	-0.56448	0.42791	-1.32	0.2069	1.96410
x2	1	-0.51315	0.22449	-2.29	0.0372	2.38821
x3	1	-0.07196	0.45457	-0.16	0.8763	1.37324



2. KNNL Problem 9.14 - If multicollinearity is a problem, adjustments should be made prior to selecting the best three hierarchical models. Typically you'd assess the model assumptions prior to your final decision. In this case, just use the model selection criteria to choose the three. The next problem deals with checking other model assumptions.

   Since we're asked to use the centered values, there is no multicollinearity problem. Below are the 3 best hierarchical models. This refers to chosing models such that if a higher order term like the interaction or quadratic term is included then the lower order terms should also be included. There is very little difference between these three models. `x1c` means centered X_1, `x1csq` means the square of centered X_1, etc.



	R-square	AdjRq	Model Parameters
4	0.8061	0.7507	x1c x2c x1csq x2csq
3	0.7922	0.7507	x1c x2c x1cx2c
4	0.7990	0.7416	x1c x2c x1cx2c x2csq

3. KNNL Problem 9.23

   Here I will focus on the model with three predictors, i.e., x1c, x2c, and x1cx2c. Its PRESS statistic has value 5102.49, which is much larger than the corresponding SSE value 1680.46. Therefore, it is too optimistic to take MSE as an indicator of the predictive ability of the fitted model.

   Since case 8 alone accounts for about one-half of the entire PRESS statistics, we may simply remove case 8 from the data. As will be discussed in Chapter 10, we can find out that case 8 has a large value of Cook's D and also large 'DF' statistics values (such as DFFITS, DFBETAS). We may also take a robust regression to lessen the effects of case 8, or even explore a better model.

4. KNNL Problem 9.24

   For $X_i = 10$: $[E\hat{Y} - EY]^2 = (-55)^2$, $[\hat{Y} - E\hat{Y}]^2 = (-47)^2$ where $EY = 15 + 20 * X_i + 3 * X_i^2$, $E\hat{Y} = Eb_0 + E(b_1) * X_i$, $\hat{Y} = 13 + 40 * X_i$.

   Similarly for $X_i = 20$: $[E\hat{Y} - EY]^2 = (-705)^2$, $[\hat{Y} - E\hat{Y}]^2 = (-97)^2$

5. KNNL Problem 10.20.

   It is unclear here whether you are to use the centered or uncentered variables. Here I only present the results using centered variables. When uncentered variables are used, many of the results are the same but some (the influence summaries) are slightly different.

   First, I give the model output for the transformed model. The four residual plots are then shown. There does not appear to be any further changes to the model based on these except to notice that the variability does seem to get smaller as x2 increases. This may be due to some influential points.

Analysis of Variance

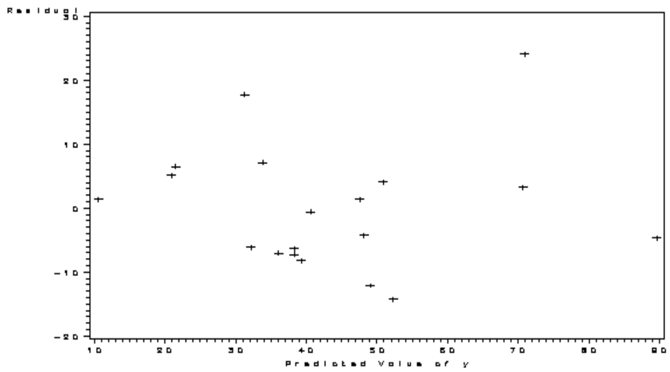
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	6407.21941	2135.73980	19.06	<.0001

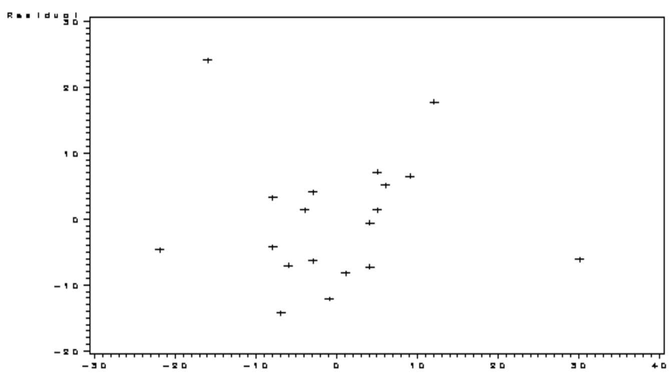
Error	15	1680.46480	112.03099
Corrected Total	18	8087.68421	

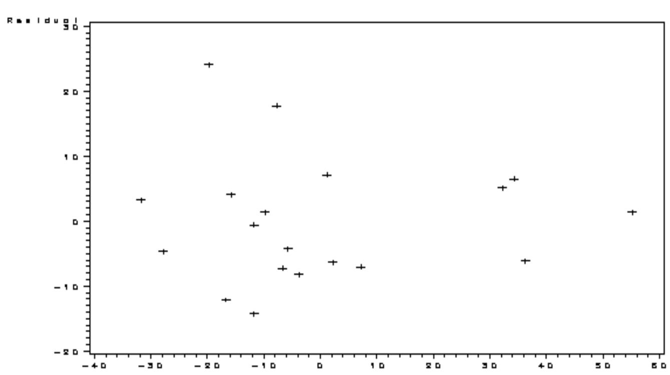
Root MSE	10.58447	R-Square	0.7922
Dependent Mean	43.26316	Adj R-Sq	0.7507
Coeff Var	24.46532		

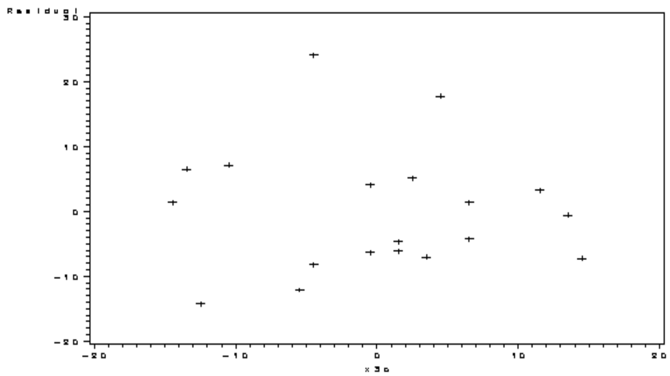
Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	1	37.88818	2.85220	13.28	<.0001	0
x1c	1	−0.67453	0.29678	−2.27	0.0382	1.75458
x2c	1	−0.60239	0.14388	−4.19	0.0008	1.82193
x1cx2c	1	0.03335	0.00928	3.59	0.0027	1.13073



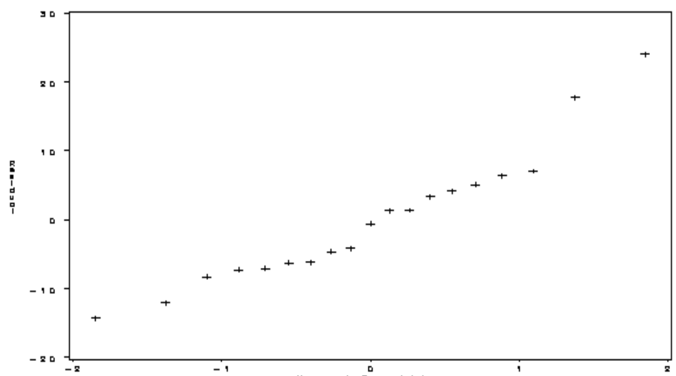






It is very clear through the VIF (above) that the centering takes care of multicollinearity.

The normality assumption appears reasonable. The normal qqplot is shown below. The correlation between the expected values and the ordered values is 0.96338. This means there is not enough evidence to reject the null.



The studentized deleted residuals are shown below. They are the same for both models. Since there are 19 observations, we'd reject if the abs(residual) were larger than 3.649. This is based on 15-1=14 degrees of freedom and the Bonferonni correction.

The hat values are also shown below. These are also the same in both models. The rule of thumb is to see if any value is greater than 2(4)/19=0.421. In this case, observation 3, 8, and 15 appear influential.

The DFbetas are also shown below. In looking whether there are values greater than 1, we see that observation 3 is ok, observation 7 has some impact on x1c, observation 8 has impact on x1c, x2c, and x1cx2c, and observation 15 is ok. In the untransformed case, observation 3 looks ok, observation 7 influences both the intercept and x1. Observation 8 influences all the parameters and observation 15 is ok.

The DFFITS and Cook's D is not shown but for observation 8, the values are -4.7798 and 4.991 respectively. These suggest concern.

```
                    Hat Diag     Cov           ------------DFBETAS-------------
Obs Residual RStudent       H   Ratio   DFFITS Intercept     x1c       x2c   x1cx2c
1  17.7397    2.2095  0.2757  0.5499   1.3632  0.8195   1.0978  -0.6979  -0.5955
```

2	4.1605	0.3989	0.0834	1.3741	0.1203	0.0923	0.0248	-0.0594	-0.0209
3	-4.6164	-0.6292	0.5389	2.5561	-0.6802	0.0718	0.3822	0.0726	-0.4819
4	-6.2590	-0.6049	0.0848	1.2988	-0.1842	-0.1722	0.0558	-0.0706	0.0929
5	5.0963	0.5172	0.1757	1.4821	0.2387	0.1334	-0.0587	0.1857	-0.0422
6	6.4898	0.6618	0.1737	1.4100	0.3035	0.1367	-0.0320	0.2039	0.0104
7	24.0398	3.3141	0.2178	0.1661	1.7486	0.2873	-1.0884	-0.0756	0.8475
8	-6.1424	-1.7794	0.8783	4.7895	-4.7798	0.7275	-2.4073	1.0195	-3.2858
9	3.3135	0.3379	0.1925	1.5799	0.1650	0.0367	0.0148	-0.1168	0.0701
10	-12.0731	-1.2232	0.1017	0.9773	-0.4116	-0.3054	-0.1630	0.2391	0.1017
11	-7.2550	-0.7153	0.1116	1.2849	-0.2534	-0.2103	-0.1407	0.0974	0.1183
12	1.3548	0.1282	0.0680	1.4073	0.0346	0.0309	-0.0024	-0.0056	-0.0095
13	-14.3075	-1.4574	0.0753	0.8100	-0.4159	-0.3130	0.1345	0.0381	0.0323
14	7.1013	0.6921	0.0929	1.2699	0.2215	0.1995	0.1044	-0.0314	-0.1097
15	1.4369	0.1821	0.4798	2.5095	0.1749	0.0575	-0.0835	0.1617	-0.0157
16	-4.1795	-0.4021	0.0897	1.3826	-0.1262	-0.0972	0.0694	-0.0338	0.0283
17	-7.0614	-0.7092	0.1444	1.3375	-0.2914	-0.2245	0.1624	-0.1828	0.1426
18	-0.5824	-0.0573	0.1391	1.5292	-0.0230	-0.0172	-0.0141	0.0119	0.0099
19	-8.2559	-0.8021	0.0768	1.1926	-0.2314	-0.2230	-0.0620	0.0332	0.1143

```
Sum of Residuals                          0
Sum of Squared Residuals         1680.46480
Predicted Residual SS (PRESS)    5102.49435
```

Output Statistics

			Hat Diag	Cov		DFBETAS			
Obs	Residual	RStudent	H	Ratio	DFFITS	Intercept	x1	x2	x1x2
1	17.7397	2.2095	0.2757	0.5499	1.3632	-0.7472	1.0870	0.2239	-0.5955
2	4.1605	0.3989	0.0834	1.3741	0.1203	0.0072	0.0304	-0.0059	-0.0209
3	-4.6164	-0.6292	0.5389	2.5561	-0.6802	-0.6519	0.5919	0.4334	-0.4819
4	-6.2590	-0.6049	0.0848	1.2988	-0.1842	0.0406	-0.0405	-0.1059	0.0929
5	5.0963	0.5172	0.1757	1.4821	0.2387	-0.0487	-0.0006	0.1089	-0.0422
6	6.4898	0.6618	0.1737	1.4100	0.3035	-0.0276	-0.0263	0.0719	0.0104
7	24.0398	3.3141	0.2178	0.1661	1.7486	1.4541	-1.2776	-0.7415	0.8475
8	-6.1424	-1.7794	0.8783	4.7895	-4.7798	-1.5469	1.1866	3.1623	-3.2858
9	3.3135	0.3379	0.1925	1.5799	0.1650	0.1021	-0.0461	-0.1051	0.0701
10	-12.0731	-1.2232	0.1017	0.9773	-0.4116	0.0359	-0.1717	0.0092	0.1017
11	-7.2550	-0.7153	0.1116	1.2849	-0.2534	0.1089	-0.1720	-0.0608	0.1183
12	1.3548	0.1282	0.0680	1.4073	0.0346	0.0013	0.0060	0.0058	-0.0095
13	-14.3075	-1.4574	0.0753	0.8100	-0.4159	-0.1260	0.0513	-0.0120	0.0323
14	7.1013	0.6921	0.0929	1.2699	0.2215	-0.1075	0.1446	0.0797	-0.1097
15	1.4369	0.1821	0.4798	2.5095	0.1749	-0.0155	-0.0353	0.0771	-0.0157
16	-4.1795	-0.4021	0.0897	1.3826	-0.1262	-0.0227	0.0174	-0.0372	0.0283
17	-7.0614	-0.7092	0.1444	1.3375	-0.2914	0.0519	-0.0186	-0.1920	0.1426
18	-0.5824	-0.0573	0.1391	1.5292	-0.0230	0.0091	-0.0157	-0.0036	0.0099
19	-8.2559	-0.8021	0.0768	1.1926	-0.2314	0.0806	-0.1241	-0.0828	0.1143

```
Sum of Residuals                          0
Sum of Squared Residuals         1680.46480
Predicted Residual SS (PRESS)    5102.49435
```

6. KNNL Problem 10.24.

$$
\begin{aligned}
H &= X(X'X)^{-1}X' \\
&= XX^{-1}(X')^{-1}X' \\
&= I
\end{aligned}
$$

This means $h_{ii} = 1$ and $\hat{Y}_i = Y_i$.

7. KNNL Problem 10.26.

Denote $\hat{Y} = (\hat{Y}_1, \cdots, \hat{Y}_n)^T$. Then $var(\hat{Y}) = \sigma^2 H$. Therefore, $\sum_{i=1}^n \sigma^2\{\hat{Y}_i\} = \sigma^2 \text{trace}(H) = \sigma^2 \sum_{i=1}^n h_{ii} = p\sigma^2$.

```
/* SAS Codes for Q1 */
data hw6q1;
    infile 'U:\.www\datasets525\ch09pr13.txt';
    input y x1 x2 x3;

proc univariate data=hw6q1;
    var x1 x2 x3;
    histogram x1 x2 x3 / normal;

proc reg data=hw6q1;
    model y= x1 x2 x3 / vif;
    output out=q1out r=res p=pred;

proc gplot data=q1out;
    plot res*pred;
run; quit;

/* SAS Codes for Q2 */
data hw6q2; set hw6q1;
 x1c=x1-32.89474;  x2c=x2-43.73684;  x3c=x3-40.47368;
 x1csq = x1c*x1c;  x2csq = x2c*x2c;  x3csq = x3c*x3c;
 x1cx2c = x1c*x2c; x1cx3c = x1c*x3c; x2cx3c = x2c*x3c;
 x1x2 = x1*x2;

proc corr data=hw6q2;
    var y x1c x2c x1cx2c;

proc reg data=hw6q2;
    model y = x1c x2c x3c x1cx2c x1cx3c x2cx3c x1csq x2csq x3csq /
              selection=rsquare adjrsq best=5;
run; quit;

/* SAS Codes for Q3 */
proc reg data=hw6q2 outest=sumstatsq3 press;
model y = x1c x2c x1cx2c/r influence;
run; quit;

proc print data=sumstatsq3; run; quit;

/* SAS Codes for Q4 */
proc reg data=hw6q2;
    model y = x1c x2c x1cx2c / influence r vif;
    output out=q4out r=res p=pred;

proc gplot data=q4out;
    plot res*pred res*x1c res*x2c res*x3c;

proc univariate noprint data=q4out;
    qqplot res / normal;

proc rank data=q4out normal=blom out=q4out2;
    var res;
    ranks normscore;

proc corr;
    var normscore res;
run; quit;
```